

SORTMATE

Team Name: AutoSparks

AI- Powered WASTE MANAGEMENT and SUSTAINABILITY PLATFORM

Open Innovation : Built Beyond the Boundaries



INTRODUCTION - Smart waste Segregation: A Step Towards Sustainable Living

PROBLEM

- Low recycling efficiency.
- Increased landfill burden.
- Environmental pollution.

POPULATION

- Rapid Urbanization has led to a significant increase in waste generation.
- 70-80% households do not segregate waste at source.

SOLUTION

- There is a need for a smart, automated and user friendly solution at household level.
- SortMate.

PROBLEMS OUR BUSINESS ADDRESSES

The waste recovery ecosystem is fragmented, causing value to be lost at every stage:

Waste Generators

Incorrect segregation and lack of awareness about the right recovery pathway leads to recyclable material being contaminated or discarded.

Waste Pickers & Recovery Partners

Difficulty finding **consistent, correctly segregated waste**, suitable sources and predictable pickup opportunities reduces their recovery efficiency and earnings.

Brands

Brands need to meet **EPR and sustainability goals**, but often lack a transparent, traceable system to track material from collection to actual recovery and generate reliable recovery data.

THE GAP

Waste is identified → but not necessarily recovered.

There is no seamless system connecting **correct segregation → recovery partners → collection → verified recovery → traceable impact.**

Waste is generated every day, but a large portion of potentially recoverable material is **incorrectly sorted, poorly tracked, or never reaches the right recovery partner.**

Current waste-management systems often focus on collection or disposal, while the complete journey from **waste identification to verified recovery** remains fragmented.

The problem is not simply identifying waste — it is ensuring that correctly identified waste actually moves through the right recovery chain and can be traced to verified recovery.

SOLUTION

① IDENTIFY & VERIFY

AI-Powered Waste Classification

Upload / capture waste image

→ AI identifies the material and suggests the correct waste category.

AI Analysis

- Waste type & material category
- Confidence score
- Explanation for the classification

Human-in-the-Loop

- User confirms AI result
- Or corrects the classification
- Verified result becomes the basis for recovery

Output:

 **Verified Waste** →
Recovery-Ready

② MATCH & MOVE

AI-Assisted Recovery Partner Matching

 **Create a Waste Batch-** Material type, Quantity, Source/location, Batch ID

Smart Partner Matching

Matches the batch with a suitable:

- Waste picker
- Aggregator
- Recycler

Based on:

- Material compatibility
- Capacity
- Minimum quantity
- Distance
- Availability
- Partner verification

 **Pickup Management** Schedule pickup, Partner accepts-Collection is tracked

Output:

 **Waste** → **Right Recovery Partner**

③ TRACE & MEASURE

End-to-End Waste Journey & Impact

 Track every stage

Created → **Verified** → **Matched** → **Pickup Scheduled** → **Collected** → **Sent to Recycler** → **Recovered**

Traceability

- Batch & pickup ID
- Partner details
- Timestamps
- Recovery status
- Material quantity

For Brands & Institutions

- Verified recovery quantities
- Material-wise recovery data
- Partner/recycler traceability
- Sustainability & EPR-support reporting
- Estimated environmental impact

Output:

 **From Waste** → **Verified Recovery** →
Measurable Impact

PROTOTYPE

HOW IT WORKS?

1. Identify

- User uploads/captures a waste image.
- AI analyzes the image and suggests a waste category.
- Confidence level and explanation are displayed.

2. Verify

- User confirms or corrects the AI result.
- The verified classification is stored with timestamp and confidence.

3. Create Waste Batch

- Verified recyclable/e-waste/dry material can become a recovery batch.
- Batch stores material, quantity, source and status.

4. Match

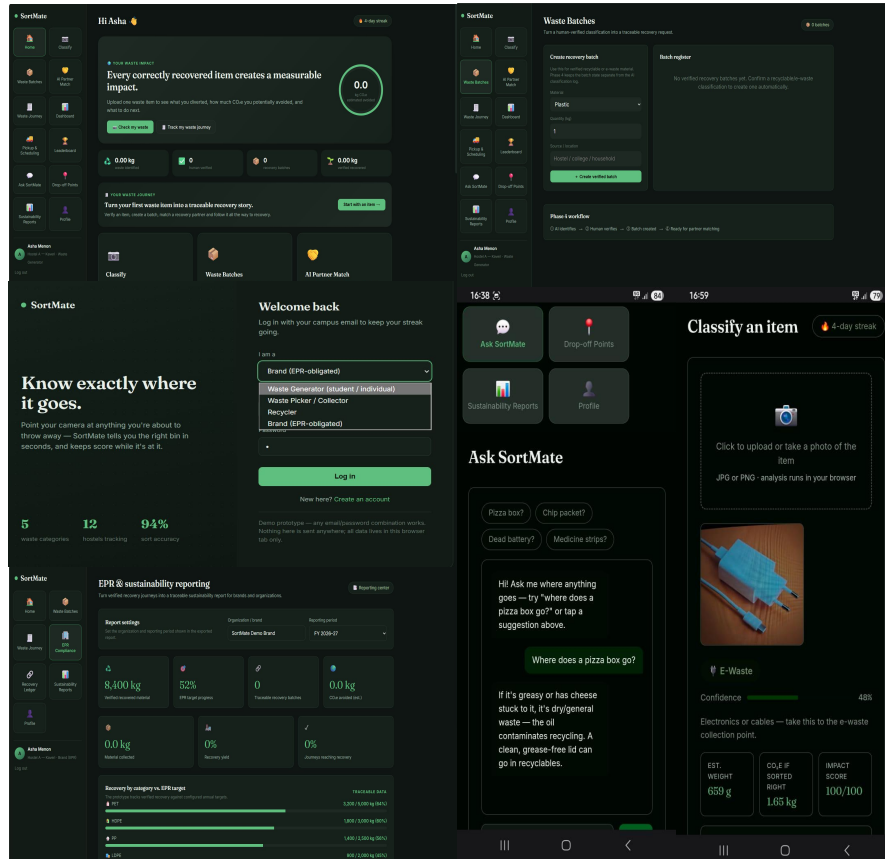
- SortMate scores suitable recovery partners using:
material compatibility + capacity + quantity + distance + availability + verification.

5. Move

- User schedules pickup.
- Partner accepts and collection status is updated.

6. Track & Measure

- The Waste Journey tracks the complete movement of a waste batch.
- Dashboard calculates recovery and impact metrics from stored prototype data.



USERS / ECOSYSTEM



WASTE GENERATOR

Households • Hostels
Colleges • Offices



WASTE PICKER / AGGREGATOR

Collection • Sorting
Aggregation



RECYCLER

Material Recovery
& Processing



BRAND / INSTITUTION

EPR Compliance
& Sustainability Reporting

AI & DECISION LAYER

1

AI VISION ENGINE



- Image-based identification
- Category prediction
- Confidence score
- Explanation



2

HUMAN-IN-THE-LOOP



- Confirm AI result
- Correct classification
- Verified waste category



3

AI-ASSISTED MATCHING ENGINE



- Material compatibility
- Quantity & capacity
- Distance & availability
- Partner verification

PLATFORM LAYER



WASTE BATCH MANAGEMENT

- Material type • Quantity
- Source • Batch ID



PICKUP & JOURNEY ENGINE

- Schedule pickup
- Track collection status
- Update journey



RECOVERY & IMPACT TRACKING

- Recovery confirmation
- Material flow tracking
- Impact calculation



EPR & REPORTING

- Recovery quantities
- Compliance reports
- Sustainability metrics

FRONTEND & DATA LAYER



Browser Local Storage

- Stores user data
- Saves batches & journey
- Prototype data persistence



JavaScript (ES6+)

- State management
- Dynamic UI & interactions
- API integration (AI vision ready)



HTML + CSS

- Responsive UI
- Clean & simple interface
- Single-page application



No Backend Used

Entire platform runs on the
frontend with local storage
(for prototype/demo).

TECH STACK



HTML
(structure)



CSS
(styling)



JavaScript
(interactivity)



LocalStorage
(data persistence)



No Backend
(fully frontend-based)



From Waste Identification to Verified Recovery — All in One Platform.



REVENUE GENERATION & BUSINESS MODEL

B2B / B2B2C Waste Recovery Platform

1. 🌱 Recovery Management — Subscription

- Colleges • Offices • RWAs • Hotels • Institutions
- Monthly fee for waste tracking, pickup management & dashboard

2. 💰 Recovery Transaction Fee

- Fee on successfully completed waste-recovery transactions
- Connects generators with pickers, aggregators & recyclers

3. 🏢 Brand & EPR Reporting

- Brands pay for traceable recovery data
- Material-wise recovery reports & sustainability/EPR-support analytics

4. 🚚 Logistics Coordination

- Service/coordination margin on scheduled waste pickups

Revenue Flow



Business Model

Recurring SaaS + Recovery Transaction Fees + Brand Reporting + Logistics Services

Why We Stand Out

Not just waste identification — a complete recovery ecosystem.

1 Smarter Identification

AI-powered waste classification with human verification.

Plastic (PET)
Confidence: 92%

2 Direct Recovery Network

AI matches waste with the right pickers, aggregators or recyclers.

Pickers
Aggregators
Recyclers

3 Transparent & Measurable

End-to-end tracking with verified recovery data and EPR support for brands.

Created → Collected → Recycled
Recovery Report
EPR

The only platform that connects people, partners and data for real recovery and lasting impact.

Market Scalability

GEOGRAPHIC EXPANSION

One city → Multiple cities → **Pan-India recovery network**



TECHNOLOGY & WASTE EXPANSION

One platform scales across **users, pickups, institutions and waste batches**, expanding from **Plastic, Paper & E-waste** → **Glass, Metal, Organic & Textile**.



PARTNER & ENTERPRISE NETWORK

Scale through more **Pickers, Aggregators, Recyclers** and customers from **Colleges** → **Corporates** → **Brands**.



DATA & ANALYTICS GROWTH

Growing recovery data enables better **AI matching, material analytics, partner performance, sustainability metrics and EPR reports**.



GO-TO-MARKET STRATEGY

PHASE 1: PILOT LAUNCH

- Test in hostels/apartments/colleges
- Gather feedback and improve product

PHASE 2: EARLY ADOPTION

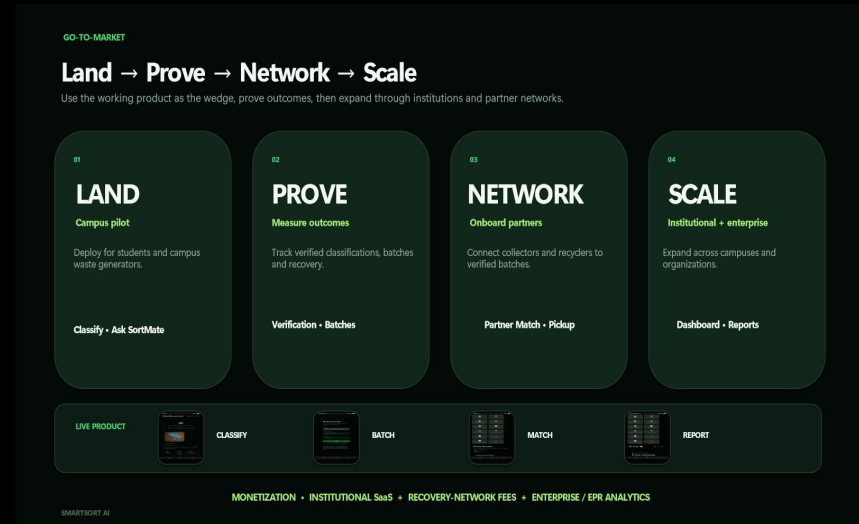
- Target housing societies and institutions
- Offer demo+bulking pricing

PHASE 3: PARTNERSHIP

- Collaborate with municipalities and waste management companies
- Leverage SMART CITY and sustainability initiatives

PHASE 4: BRAND AWARENESS

- Social media campaigns
- Sustainability driven content and workshop.



Key Challenges



Accuracy at the edge

Mixed items, low light, and messy real-world photos are harder than clean demo images — the model must generalize far beyond a curated dataset



Cold-start network effects

AI matching only works once enough pickers, aggregators, and recyclers are active — an empty marketplace helps no one.



Ground-level connectivity

Waste pickers often use basic phones with patchy data — the picker flow must work reliably offline-first.



Gaming & data trust

Photo self-reporting for points can be spoofed. Anti-fraud checks and spot audits are essential before rewards carry real value.

Future Roadmap

Phase 1 • Foundation

0–3 months

- ▶ Persistent cloud backend, replacing local-only demo state
- ▶ Train the classifier on a real, labelled campus-waste image set
- ▶ UPI-based instant payouts for pickers

Phase 2 • Scale-up

3–6 months

- ▶ Expand from one campus to a city/ward-level deployment
- ▶ IoT-enabled smart bins for auto-fill & contamination alerts
- ▶ Multilingual voice support (Hindi + regional languages)

Phase 3 • Integration

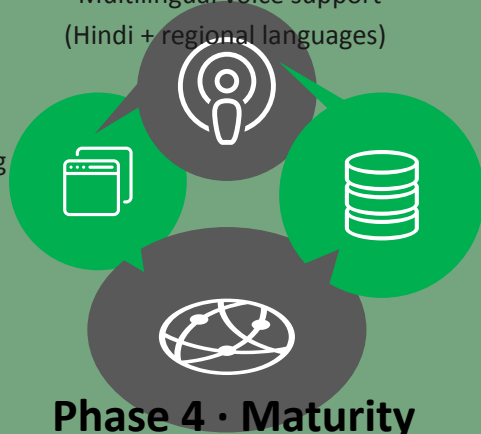
6–12 months

- ▶ Municipal / ULB pilot integration
- ▶ Tamper-evident chain-of-custody for audit-grade EPR reports
- ▶ Open API for third-party municipal & corporate integrations

Phase 4 • Maturity

12–18 months

- ▶ Predictive analytics for waste-generation forecasting
- ▶ National-scale recycler & brand marketplace





Thank You!

Presented by:

Sudiksha Agrawal

2nd BTech
Biochemical Engineering

Priyanshi Yadav

2nd BTech
Electronics Engineering

Vanshika Sinha

2nd BTech
Electronics Engineering